🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.NS.1

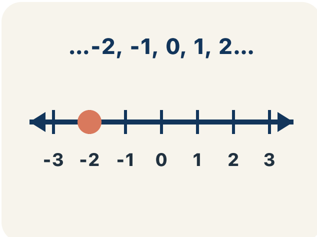
Divide Whole Numbers by Fractions

Lesson 2-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary **Level 2 Standard**

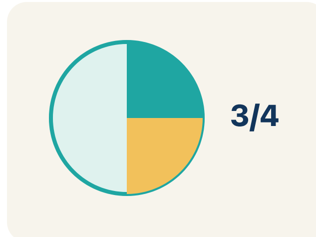
Picture first, then the word, then a plain-language meaning. Say each word out loud.



4 is a whole number; as a fraction it is 4/1

Whole Number

Write the definition:



1/2 means 1 out of 2 equal parts — like cutting a sandwich in half and taking one piece

Fraction

Write the definition:

3/4 → 4/3

flip top and bottom

The reciprocal of 2/3 is 3/2. Flip the fraction to divide: 6 ÷ 2/3 = 6 × 3/2

Reciprocal

Write the definition:

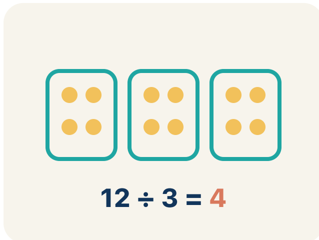
3/4 → 4/3

flip top and bottom

6 ÷ 1/3 → Keep 6, Change ÷ to ×, Flip 1/3 to 3/1 → 6 × 3 = 18

Keep, Change, Flip

Write the definition:



In 4 ÷ 1/2 = 8, the quotient is 8

Quotient

Write the definition:

Guided Notes
Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can divide a whole number by a fraction by multiplying by the reciprocal.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Whole Number | **Fraction** | **Reciprocal** | **Keep, Change, Flip** | **Quotient**

👆 *Tap any word to see what it means and a picture.*

1. A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a ____________.

2. A number that shows part of a whole, like 3/4, is a ____________.

3. The fraction you get by flipping a fraction upside down is its ____________.

4. To divide by a fraction, I ____________: keep the first number, change ÷ to ×, and flip the second fraction.

5. The answer to a division problem is the ____________.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: What is 6 ÷ 1/3?

A. 18

B. 2

C. 6/3

D. 3

Step 1 6 ÷ 1/3 = 6 × 3/1 = 18.

Step 2 There are 18 thirds in 6 wholes.

✅ **Answer:** A. 18

🧩 **Try — put the steps in order**

Drag the cards (or use the ▲▼ buttons) to put the solution steps in the right order, then press **Check**.

There are 18 thirds in 6 wholes.

6 ÷ 1/3 = 6 × 3/1 = 18.

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: What is 3 ÷ 1/5?

A. 15

B. 3/5

C. 5/3

D. 8

Step 1 ______________________

Step 2 ______________________

Answer: ______________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: What is 10 ÷ 1/4?

A. 40

B. 2.5

C. 14

D. 10/4

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. There are 3 bags of trail mix. Each snack pouch needs 2/3 of a bag. How many full pouches can be filled, and is there a leftover?

A. 4 1/2 pouches (4 full, half a pouch left)

B. 4 pouches exactly

C. 2 pouches

D. 6 pouches

Show your work:

2. Detective Park bakes with 9 cups of batter. Each muffin pan holds 3/4 of a cup. How many pans can he fill?

A. 12

B. 6

C. 27/4

D. 9

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A pizza shop has 8 whole pizzas. Each serving is 2/3 of a pizza. How many servings can they make? Write and solve an equation, then explain why the answer is greater than 8.

Sentence starter: The equation is 8 ÷ 2/3 = ____. Using Keep, Change, Flip: 8 × 3/2 = ____. The answer is greater than 8 because ____.

Show your work:

Reflect — Exit Ticket

What is 9 ÷ 1/3?

A. 27

B. 3

C. 9/3

D. 1/27

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Whole Number
2. **Notes 2:** Fraction
3. **Notes 3:** Reciprocal
4. **Notes 4:** Keep, Change, Flip
5. **Notes 5:** Quotient
6. **We Try (notes):** A. 15 *— 3 ÷ 1/5 = 3 × 5/1 = 15. There are 15 fifths in 3 wholes.*
7. **You Try (notes):** A. 40 *— 10 ÷ 1/4 = 10 × 4/1 = 40. There are 40 quarter-size pieces in 10 wholes.*
8. **Try It 1:** A. 4 1/2 pouches (4 full, half a pouch left) *— 3 ÷ 2/3 = 3 × 3/2 = 9/2 = 4 1/2. You can fill 4 full pouches with half a pouch of mix left over.*
9. **Try It 2:** A. 12 *— 9 ÷ 3/4 = 9 × 4/3 = 36/3 = 12 pans.*
10. **Exit Ticket:** A. 27 *— 9 ÷ 1/3 = 9 × 3/1 = 27. There are 27 thirds in 9 wholes.*